SMART ENVIRONMENT STATION

Software Requirements Specification (SRS)

Functional and Non-Functional Requirements

Student Project Draft | *Version 1.0*

Prepared: 12 October 2026

1. Introduction

The Smart Environment Station is an IoT-based system that monitors environmental conditions. An ESP32 microcontroller collects measurements from connected sensors, including temperature, humidity, air pressure, and light intensity, and transmits the readings over Wi-Fi to an ASP.NET Core backend. The backend validates and stores readings and provides data to a mobile application. Users can monitor current readings, view historical trends, manage devices, configure thresholds, and receive alerts when configured limits are exceeded.

This document defines proposed functional and non-functional requirements for the student project. Requirements should be reviewed against the final agreed scope and verified through testing.

2. Functional Requirements

Functional requirements specify the services and behavior the system must provide. Each requirement has a unique identifier.

ID	Requirement	Description
FR-01	Sensor Data Collection	The system shall collect temperature, humidity, air pressure, and light-intensity readings from the sensors connected to the ESP32.
FR-02	Wireless Communication	The ESP32 shall transmit sensor readings to the backend server over Wi-Fi using the communication protocol selected for the project.
FR-03	Input Validation	The backend shall validate incoming sensor readings and reject or flag invalid payloads before storing them.
FR-04	Data Storage	The system shall store each accepted sensor reading with a timestamp and the identifier of the device that produced it.
FR-05	Current Readings	The mobile application shall display the latest available environmental readings for each device the user is permitted to access.
FR-06	Historical Data	The mobile application shall allow users to select a time period and view historical sensor readings in charts.
FR-07	Device Management	The system shall allow users to view registered devices and perform the device-management actions included in the approved project scope.
FR-08	Threshold Configuration	The system shall allow users to configure threshold values for supported environmental measurements.
FR-09	Alert Generation	The backend shall generate an alert when a valid reading crosses a configured threshold.
FR-10	Alert Delivery	The system shall make generated alerts available to users through the notification mechanism implemented in the project.
FR-11	Device Status	The system shall show the latest known status of each device, based on received readings or an implemented heartbeat mechanism.
FR-12	User Authentication	If user accounts and protected features are in scope, the system shall authenticate users before granting access to those features.

3. Non-Functional Requirements

Non-functional requirements define measurable quality attributes and operating constraints. Numerical targets below are proposed acceptance targets, not measured results; the team should validate or revise them during testing.

ID	Quality Attribute	Description
NFR-01	Performance	Under normal test conditions on the local network, the backend shall return a response to a valid sensor-reading request within 2 seconds.
NFR-02	Data Freshness	Under normal local-network conditions, the mobile application shall display a newly processed reading within 3 seconds after the backend has accepted it.
NFR-03	Security	The system shall restrict protected user and device-management operations to authorized users. Credentials and secrets shall not be hard-coded into public source repositories.
NFR-04	Reliability	A temporary Wi-Fi disconnection shall not permanently stop the ESP32 firmware; the device shall attempt to reconnect and resume transmission when connectivity returns.
NFR-05	Data Integrity	Each stored reading shall be associated with the correct device identifier, measurement values, and timestamp. Invalid payloads shall not be stored as valid readings.
NFR-06	Usability	The mobile application shall present readings with clear labels and measurement units and shall distinguish unavailable data from a valid zero value.
NFR-07	Compatibility	The mobile application shall support the operating systems and minimum versions selected by the project team and documented in the final report.
NFR-08	Maintainability	The software shall separate device communication, backend business logic, database access, and mobile presentation into identifiable components.
NFR-09	Local Availability	During a demonstration, the backend shall be accessible to devices on the local network while the host computer, application process, and network remain operational; external Internet access shall not be required.
NFR-10	Resource Efficiency	The ESP32 firmware shall operate within the memory and processing limits of the selected board during continuous monitoring and transmission.